AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Credit Suisse: Clifton Data Center, NJ -- station KTEB, America/New_York
Building	OSM 818790301 Credit Suisse: Clifton Data Center
Facade gap	not applicable -- one building, no second facade
Weather record	43,619 real hourly records from KTEB
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-30 (chatter)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 1 mode change. That is 6 more than the reactive incumbent, at 0 unsafe hours against its 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	7 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	1 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	20.560	18.0	20.560	0.726

01	mechanical	no	dry-bulb	20.835	18.0	21.110	0.674
02	mechanical	no	dry-bulb	19.725	18.0	18.890	0.628
03	mechanical	no	dry-bulb	19.170	18.0	17.780	0.585
04	mechanical	no	dry-bulb	19.165	18.0	16.670	0.792
05	mechanical	yes	switch budget	17.780	18.0	16.110	0.696
06	mechanical	yes	switch budget	17.225	18.0	15.560	0.684
07	mechanical	yes	switch budget	17.505	18.0	15.560	0.932
08	mechanical	no	dry-bulb	19.165	18.0	16.670	1.584
09	mechanical	no	dry-bulb	19.725	18.0	17.220	1.635
10	mechanical	no	dry-bulb	20.280	18.0	18.330	1.516
11	mechanical	no	dry-bulb	20.000	18.0	17.220	1.420
12	mechanical	no	dry-bulb	19.720	18.0	17.220	1.133
13	mechanical	no	dry-bulb	20.275	18.0	18.330	0.894
14	mechanical	no	dry-bulb	18.330	18.0	16.110	0.942
15	mechanical	no	dry-bulb	19.160	18.0	18.330	1.044
16	mechanical	no	dry-bulb	19.165	18.0	18.330	0.931
17	FREE	yes	-	16.945	18.0	17.220	0.821
18	FREE	yes	-	17.220	18.0	16.110	0.888
19	FREE	yes	-	15.830	18.0	13.330	1.117
20	FREE	yes	-	14.445	18.0	11.670	1.162
21	FREE	yes	-	13.055	18.0	10.000	1.073
22	FREE	yes	-	11.110	18.0	8.890	0.928
23	FREE	yes	-	9.445	18.0	7.220	0.809

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]
Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.560 C would have passed. The outside air is too HUMID: the dew-point bound is 18.05 C against a 15.0 C maximum, failing by 3.05 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 20.835 C against a 18.0 C limit, so it fails by 2.835 C. It would take a limit 2.835 C higher, or a bound 2.835 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 19.725 C against a 18.0 C limit, so it fails by 1.725 C. It would take a limit 1.725 C higher, or a bound 1.725 C tighter, to change this hour.

03:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 19.170 C against a 18.0 C limit, so it fails by 1.170 C. It would take a limit 1.170 C higher, or a bound 1.170 C tighter, to change this hour.

04:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 19.165 C against a 18.0 C limit, so it fails by 1.165 C. It would take a limit 1.165 C higher, or a bound 1.165 C tighter, to change this hour.

05:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.225 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.505 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.165 C against a 18.0 C limit, so it fails by 1.165 C. It would take a limit 1.165 C higher, or a bound 1.165 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.725 C against a 18.0 C limit, so it fails by 1.725 C. It would take a limit 1.725 C higher, or a bound 1.725 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.280 C against a 18.0 C limit, so it fails by 2.280 C. It would take a limit 2.280 C higher, or a bound 2.280 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.000 C against a 18.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.720 C against a 18.0 C limit, so it fails by 1.720 C. It would take a limit 1.720 C higher, or a bound 1.720 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.275 C against a 18.0 C limit, so it fails by 2.275 C. It would take a limit 2.275 C higher, or a bound 2.275 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.330 C against a 18.0 C limit, so it fails by 0.330 C. It would take a limit 0.330 C higher, or a bound 0.330 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.160 C against a 18.0 C limit, so it fails by 1.160 C. It would take a limit 1.160 C higher, or a bound 1.160 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.165 C against a 18.0 C limit, so it fails by 1.165 C. It would take a limit 1.165 C higher, or a bound 1.165 C tighter, to change this hour.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.945 C, which is 1.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.821 C, and the margin is measured, not chosen: 0.821 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.220 C, which is 0.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.888 C, and the margin is measured, not chosen: 0.888 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.830 C, which is 2.170 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.117 C, and the margin is measured, not chosen: 1.117 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.445 C, which is 3.555 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.162 C, and the margin is measured, not chosen: 1.162 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.055 C, which is 4.945 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.073 C, and the margin is measured, not chosen: 1.073 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.110 C, which is 6.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.928 C, and the margin is measured, not chosen: 0.928 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.445 C, which is 8.555 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.809 C, and the margin is measured, not chosen: 0.809 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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